

Reproducible Tumor-Normal Dysregulation and Prognostic Discordance in Clear Cell Renal Cell Carcinoma

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Abstract—Clear cell renal cell carcinoma (ccRCC) is strongly shaped by VHL/HIF-associated hypoxia biology, but genes that distinguish tumor from normal kidney are not necessarily the same genes that carry prognostic information. This study builds a reproducible public-data pipeline to test whether tumor-normal differential expression magnitude aligns with survival-associated effect size in ccRCC. TCGA-KIRC STAR-count RNA-seq data were analyzed with DESeq2, and two independent GEO cohorts, GSE40435 and GSE53757, were analyzed with limma. Reproducibility was defined by TCGA significance, consistent direction across both GEO cohorts, and nominal validation support in at least one GEO cohort. Cox proportional hazards models were fit in TCGA primary tumors using continuous expression, stage, grade, age, and sex. The current run identified 3,304 reproducible differentially expressed genes, 519 strict reproducible prognostic candidates, and 24 high-confidence candidates after proportional-hazards, effect-size, GEO-effect, and sensitivity filtering. These results remain computational and retrospective, but the candidate set is now compact enough for focused biological interpretation rather than broad gene-list reporting. The main contribution is a transparent framework for stress-testing the assumption that tumor identity genes are also prognostic genes.

Index Terms—clear cell renal cell carcinoma, TCGA, GEO, differential expression, survival analysis, reproducibility, hypoxia, transcriptomics

I. INTRODUCTION

Clear cell renal cell carcinoma is the most common histologic subtype of kidney cancer and has a well-established molecular relationship with VHL loss, HIF stabilization, angiogenesis, metabolic rewiring, and microenvironmental adaptation [1], [2]. Public transcriptomic resources make ccRCC attractive for student-scale computational biology, but they also make it easy to produce shallow “hub gene” analyses that overinterpret differential expression as clinical importance.

The central premise of this project is deliberately narrower: tumor-normal dysregulation and prognostic relevance are related but not equivalent. A gene can be strongly differentially expressed because it marks kidney tissue identity, tumor composition, immune infiltration, or metabolic state, without independently explaining patient outcome. Conversely, a modest expression change may carry survival signal if it reflects aggressive adaptation.

This work asks: in ccRCC, do the most reproducibly dysregulated tumor-normal genes also carry prognostic information, or is survival associated with a narrower hypoxia-linked adaptation program? The goal is not to claim therapeutic targets.

The goal is to build a reproducible evidence funnel: differential expression, cross-cohort validation, adjusted survival modeling, enrichment, and a discordance figure comparing expression perturbation with survival effect.

II. METHODS

A. Datasets

TCGA-KIRC was used as the discovery cohort. RNA-seq data were downloaded from the Genomic Data Commons as STAR-count gene expression quantification using TCGAbiolinks [3]. The first run included 541 primary tumor samples and 72 solid tissue normal samples. Clinical metadata were downloaded from GDC and used to derive overall survival time and event status.

Two GEO cohorts were used for first-pass expression validation [4]. GSE40435 contained 101 adjacent non-tumor kidney samples and 101 ccRCC tumor samples. GSE53757 contained 72 matched normal kidney samples and 72 ccRCC tumor samples. GEO Series Matrix files and supplementary raw archives were downloaded by script; the current analysis uses the Series Matrix expression values for a fast validation pass.

B. Differential Expression

TCGA raw unstranded STAR counts were analyzed with DESeq2 [5]. Genes were retained if at least 10 counts were observed in at least 10 samples. The design compared primary tumor against solid tissue normal. TCGA significance was defined as Benjamini-Hochberg adjusted $p < 0.05$ and absolute $\log_2$ fold change at least 1.

GEO cohorts were analyzed with limma [6]. For GSE40435, patient-pair identifiers were parsed from sample titles. For GSE53757, paired identifiers were assigned from the documented alternating tumor-normal sample order. Probe-level expression was collapsed to gene symbols by mean expression after parsing platform gene-symbol annotations.

C. Reproducibility Definition

A gene was considered reproducibly differentially expressed if it met the TCGA differential expression threshold, had the same tumor-normal direction in both GSE40435 and GSE53757, and had nominal validation support ($p < 0.05$) in at least one GEO cohort. This definition was chosen to reduce post-hoc threshold drift while keeping enough genes for downstream survival and enrichment analysis.

D. Survival Analysis

Survival analysis used TCGA primary tumor samples only. Cox proportional hazards models [7] were fit using standardized continuous expression rather than median-split expression groups. The main model used complete cases for age, sex, pathologic stage, and collapsed grade (G1/G2 versus G3/G4). Two sensitivity models separately required stage-complete and grade-complete covariates. High-confidence candidates were required to pass main-model $FDR < 0.01$, proportional-hazards $p \geq 0.05$, absolute log hazard ratio at least $\log(1.5)$, non-trivial GEO effect support, and same-direction nominal support in both sensitivity models.

E. Enrichment and Visualization

GO Biological Process over-representation analysis was implemented directly with `org.Hs.eg.db` and `GO.db` to avoid fragile plotting dependencies. Hypergeometric tests were corrected with Benjamini-Hochberg adjustment. The signature visualization is a discordance plot with absolute tumor-normal $\log_2$ fold change on the x-axis and absolute adjusted Cox log hazard ratio on the y-axis. Point size represents survival significance, and reproducible genes are emphasized.

III. RESULTS

A. TCGA and GEO Differential Expression

The TCGA-KIRC analysis identified 14,337 Ensembl-level genes passing the initial DESeq2 threshold. After mapping to symbols, 8,852 TCGA-significant genes were available for cross-cohort comparison. GSE40435 produced 1,076 significant genes under the same effect-size threshold, and GSE53757 produced 3,139 significant genes.

The cross-cohort evidence funnel identified 3,304 reproducible differentially expressed genes. This reduction is important: the TCGA-only DEG count is large enough to be biologically diffuse, while the reproducibility filter begins to separate robust ccRCC expression programs from single-cohort findings.

TABLE I
FIRST-PASS EVIDENCE FUNNEL

Metric	Count
TCGA primary tumor samples	541
TCGA solid tissue normal samples	72
GSE40435 tumor/normal samples	101/101
GSE53757 tumor/normal samples	72/72
TCGA significant genes after symbol mapping	8,852
Same direction in both GEO cohorts	9,149
Reproducible DEGs	3,304
Strict reproducible prognostic candidates	519
High-confidence candidates	24

B. Survival Association and Discordance

Among reproducible DEGs, 1,176 were associated with overall survival in the main stage/grade-complete Cox model at $FDR < 0.05$. Additional proportional-hazards, effect-size, GEO-effect, and sensitivity filters reduced this to 519 strict

candidates and 24 high-confidence candidates. The high-confidence set is small enough for candidate-level review while preserving the full audit trail. The result supports the project premise: final interpretation must prioritize effect size, proportional hazards behavior, clinical covariate robustness, validation magnitude, and pathway coherence rather than FDR alone.

Top high-confidence candidates included CYFIP2, CRYL1, HHLA2, DDC, KL, PANK1, IQGAP2, and ACADM, many of which were downregulated in tumor and associated with lower hazard when expression was higher. Some genes had strong TCGA effects but weaker GEO fold changes, emphasizing why the final project should rank genes by cross-cohort evidence rather than a single statistic.

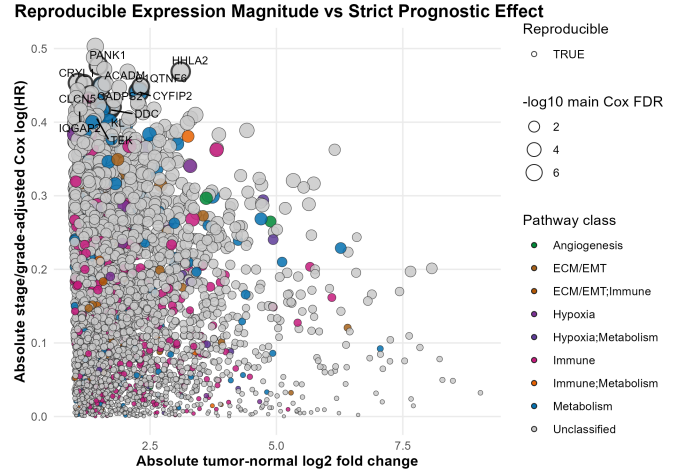


Fig. 1. First-pass discordance plot comparing absolute tumor-normal expression magnitude with absolute adjusted Cox log hazard ratio. Reproducible genes are emphasized. This figure is intended to test whether tumor-normal dysregulation magnitude aligns with prognostic effect size.

C. Pathway Enrichment

GO Biological Process enrichment of the strict candidate set and hallmark-class annotation of the high-confidence set highlighted metabolism-dominant candidates with a smaller immune component. This supports a more specific interpretation than the initial broad hypoxia framing: the current evidence points toward hypoxia-linked metabolic adaptation plus immune/microenvironmental signals. That finding is biologically plausible in ccRCC, but it should be interpreted cautiously because bulk RNA-seq cannot distinguish tumor-cell intrinsic programs from immune or stromal composition.

IV. DISCUSSION

This first implementation supports the feasibility of the project design. The analysis moves beyond a single-cohort DEG list by requiring independent GEO direction support and by comparing reproducible tumor-normal dysregulation against adjusted survival effect size. The current result also shows why restraint is necessary: thousands of genes pass early filters, but the high-confidence evidence layer compresses the project to a small reviewable set.

TCGA-KIRC Tumor vs Normal DESeq2

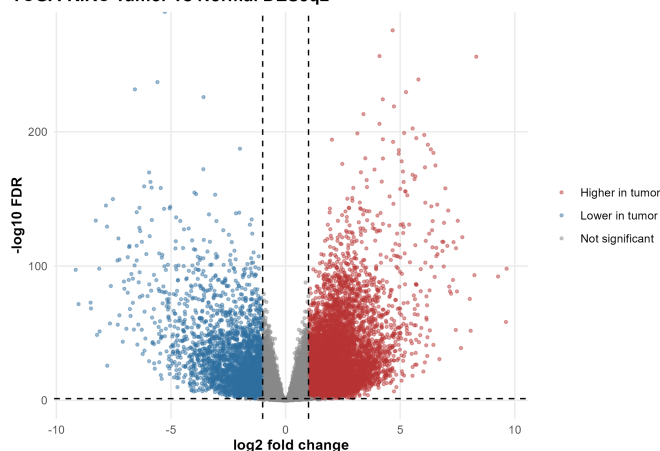


Fig. 2. TCGA-KIRC tumor-normal DESeq2 volcano plot. This plot shows the breadth of tumor-normal transcriptional disruption before cross-cohort validation and survival filtering.

The most important scientific next step is not to add machine learning. It is candidate-level biological review: each high-confidence gene should be checked against ccRCC biology, kidney cell-type specificity, and independent literature before being discussed as a candidate prognostic association.

The enrichment results currently emphasize metabolic candidates with a smaller immune component. That may reflect genuine ccRCC adaptation biology, but it may also reflect bulk tissue composition. This is precisely the kind of ambiguity the project should preserve rather than hide. Because metabolism currently dominates the annotated high-confidence set more than hypoxia itself, the final interpretation should be revised toward hypoxia-linked metabolic adaptation rather than a generic hypoxia-only claim.

V. LIMITATIONS

This analysis is retrospective and uses public bulk transcriptomic data. Adjacent normal kidney is not identical to healthy kidney. Cox associations do not prove causality. GEO validation currently uses Series Matrix expression values rather than fully reprocessed raw arrays. Clinical covariate completeness varies across TCGA samples. No wet-lab validation is included. Therefore, all gene-level findings should be framed as candidate prognostic associations, not therapeutic targets.

VI. CONCLUSION

HYDRA-ccRCC now has a complete reproducible first-pass pipeline for testing discordance between tumor-normal differential expression and prognostic survival signal in ccRCC. The current analysis identifies 3,304 reproducible DEGs, 519 strict reproducible prognostic candidates, and 24 high-confidence candidates. The main contribution at this stage is not a final biomarker list, but a transparent framework for deciding which findings deserve deeper biological interpretation.

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